Lecture 1: Linear Systems

MATH 2308

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Systems of equations

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The simplest example of a system of linear equations is a system of two equations and two variables, such as

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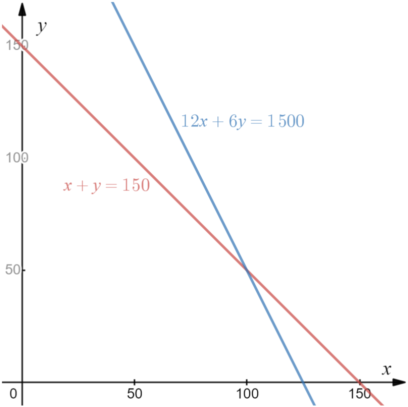
This system of equations is **linear** because both equations in the system are linear. A **solution** to the system of equations is an (x, y)-pair that satisfies both equations.

The geometry

A single linear equation in two variables has a line as its graph. A linear system of two equations in two variables involves two lines.



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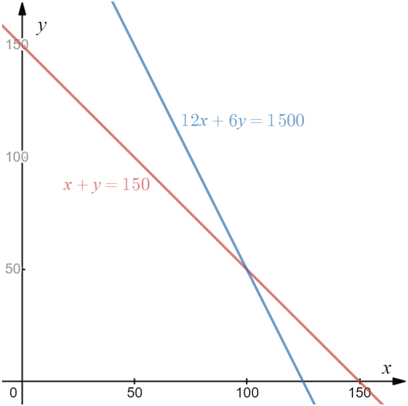
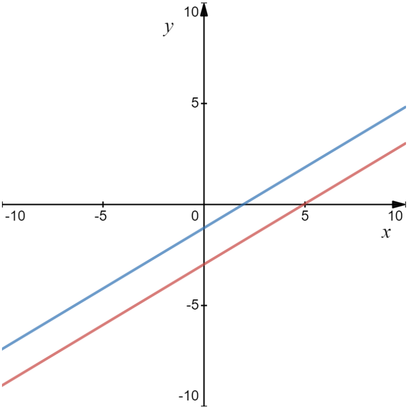


A **solution** to the system must lie on both lines. In this case, it is the point of intersection.

Not all systems

Not all systems of two linear equations in two variables have a solution. If the two lines are parallel, there is no solution.



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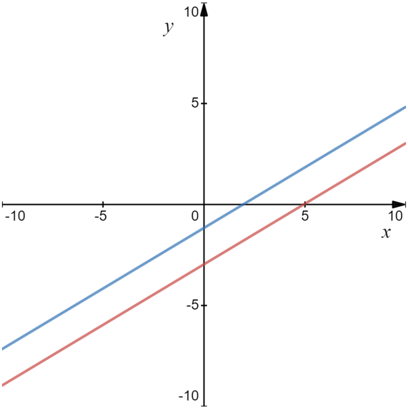
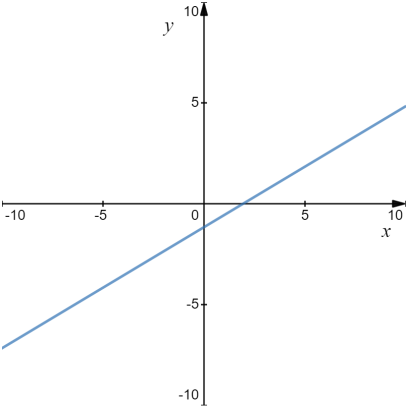


Such systems are called **inconsistent**. (A system with a solution is **consistent**.)

Dependent systems

Sometimes the two equations actually represent the same line. In that case the system is called **dependent**.



Since every point on the line is a solution to both equations, we have infinitely many solutions.

Three cases

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It turns out, when we increase the number of variables and/or equations, there are still these three cases.

Methods of solution: Gaussian Elimination

Let's return to our original system of equations and try to solve it.

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The method of **Gaussian elimination** involves getting the system into what is called **upper triangular form**

$$\begin{cases} *x + *y = * \\ *y = * \end{cases}$$

where the asterisks represent different constants.

We can put a system of equations into upper triangular form using a sequence of operations on the equations. The following operations do not change the solution set for a linear system.

- Multiplying an equation by a nonzero constant

Gaussian Elimination, continued

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Once the system is in upper triangular form, we can use **back substitution** to solve for all the variables.

Solving the system

In the system

$$\begin{cases} x + y = 150 \\ 12x + 6y = 1500 \end{cases},$$

we want to turn the x coefficient in the second equation into a 0. We can do this by multiplying the first equation by -12 and adding it to the second one.

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We can now solve for y : $y = \frac{-300}{-6} = 50$. Plugging this value in for y in the first equation gives us $x + 50 = 150$ or $x = 100$. Our solution is $(x, y) = (100, 50)$.

Verification

Let's verify that $(x, y) = (100, 50)$ satisfies the system

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For the second equation,

$$12x + 6y = 12 \cdot 100 + 6 \cdot 50 = 1200 + 300 = 1500\checkmark.$$

We have our solution. It's the only one, because the two lines are not parallel.

Another example

Solve

$$\begin{cases} x + 2y = 3 \\ 2x - 3y = 10 \end{cases}.$$

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Thus, $y = -4/7$ and $x = 3 - 2\left(-\frac{4}{7}\right) = \frac{21}{7} + \frac{8}{7} = \frac{29}{7}$.

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Since $0 \neq -5$, this equation has no solution (it is inconsistent.) The original lines were parallel.

What else?

If we tweak the last system a little bit. . .

$$\begin{cases} 2x + 3y = 6 \\ 4x + 6y = 12 \end{cases},$$

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we see that by multiplying the top equation by 2 gives us identical equations. So

- the two equations represent the same line, and
- every point on that line is a solution, and
- there are infinitely many solutions.

Parameterizing solutions

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For instance, in the last example, whose solution was the line $2x + 3y = 6$, we can let $y = t$ and solve for $x : 2x + 3t = 6$ or $x = 3 - \frac{3}{2}t$, and our solution is

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$$\left\{(3 - \frac{3}{2}t, t), t \in \mathbb{R}\right\}.$$

Such parameterizations are not unique. Letting $x = t$ gives us a different-looking solution:

$$\left\{(t, 2 - \frac{2}{3}t), t \in \mathbb{R}\right\}.$$

More equations, more variables

Here's a system of 3 linear equations in 3 variables.

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A single linear equation of the form

$$ax + by + cz = d,$$

where a, b, and c are constants (and not all zero), represents a plane in three dimensions. A solution to this system of equations would lie on the intersection of three planes. Again, the possibilities are zero, one, or infinitely many solutions.

We can again use Gaussian elimination to get the system into upper triangular form. It works best if the x-coefficient in the first equation is a 1.

$$\begin{cases} 2x - 2y + \ \ z = 12 \\ \ \ x + \ \ y - 2z = -1 \\ 3x + \ \ y + 2z = 15 \end{cases}.$$

If we interchange the first and second equations, we make that happen. (I'm dropping the left brace for lazy typesetting reasons.)

$$\begin{aligned} x + y - 2z &= -1 \\ 2x - 2y + z &= 12 \\ 3x + y + 2z &= 15 \end{aligned}$$

Gaussian elimination, continued

We can use the x term in the first equation as a "pivot" to eliminate the x term from the other two equations.

$$\begin{aligned} x + y - 2z &= -1 \\ 2x - 2y + z &= 12 \\ 3x + y + 2z &= 15 \end{aligned}$$

Replace Equation 2 with $-2\times$ Equation 1 $+$ Equation 2 ($-2E1 + E2 \to E2$), and replace Equation 3 with $-3\times$ Equation 1 $+$ Equation 3 ($-3E1 + E3 \to E3$).

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We are now in "triangular" form, and we can solve immediately for $z : z = 2$.

Finishing up: back substitution

Now that we know the value of z, we can use it in the second equation to solve for y.

$$\begin{aligned} x + y - 2z &= -1 \\ y - 4z &= -9 \\ z &= 2 \end{aligned}$$

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Our solution is $(x, y, z) = (4, -1, 2)$.

No solutions

What would a no-solution case look like? After Gaussian elimination, perhaps we end up with something like this:

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Similar to the "2×2" case, a bottom row with $0 = a$, where $a \neq 0$ would indicate an inconsistent system.

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$$\begin{aligned} y - 3t &= 1 \\ y &= 1 + 3t. \end{aligned}$$

Infinitely many solutions

Now that we have z and y in terms of the parameter t, we can solve for x using the first equation.

$$\begin{aligned} x + 2y + 4z &= 2 \\ x &= 2 - 2y - 4z \\ &= 2 - 2(1 + 3t) - 4t \\ &= -10t. \end{aligned}$$

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Our solution is

$$\{(-10t, 1 + 3t, t), t \in \mathbb{R}\}.$$

Again, this parameterization is not unique. Our solution is a line in 3 dimensions.

Another one

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With $y = s$ and $z = t$, we get $x = 2 - 4y + 3z = 2 - 4s + 3t$. Our solution is

$$\left\{(2 - 4s + 3t, s, t), (s, t) \in \mathbb{R}^2\right\}.$$

Overdetermined and underdetermined

We don't have to have the same number of equations as variables. When we have more equations than variables, the system is called **overdetermined**. For example,

$$\begin{cases} x + 2y = 3 \\ 2x - y = 5 \\ 3x + y = 8 \end{cases}$$

has three equations but only two unknowns.

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Though the above example does have a single solution, overdetermined systems often have no solution. They show up in linear regression and in other cases where there are more observations than there are variables.

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Underdetermined systems don't have to have a solution, but if there's one solution, there are infinitely many. Why?